

Low Level Design Document

Introduction

This Low Level Design (LLD) document outlines the core implementation details for **DataSnap - Visual Data Insights**, a web-based tool enabling users to upload CSV files and instantly view simple visualizations (bar and pie charts). The project demonstrates backend data processing and frontend chart rendering using Spring Boot and web technologies.

1. System Components

Component	Technology	Key Responsibilities
Frontend	HTML/JS/CSS	File upload UI, chart rendering, API integration
Backend API	Spring Boot	File upload handling, CSV parsing, data aggregation
Visualization Lib	Chart.js (JS)	Render bar/pie charts from processed data

2. Class/Interface Overview

Class/Interface	Responsibility	Key Methods/Attributes
<code>FileUploadController</code>	REST endpoint for file upload	<code>uploadFile(MultipartFile)</code>
<code>CSVProcessor</code>	Parse/process CSV data	<code>parse(InputStream)</code> , <code>aggregate()</code>
<code>ChartData</code>	Data model for chart output	<code>labels: List<String></code> , <code>values: List<Number></code>
<code>ErrorResponse</code>	Standard error response	<code>message: String</code> , <code>code: int</code>

Relationships:

- `FileUploadController` uses `CSVProcessor` to process files and returns `ChartData` to the frontend.

3. Data Structure Overview

Model	Fields	Description
<code>ChartData</code>	<code>labels: List<String></code> , <code>values: List<Number></code>	Aggregated data for chart rendering
<code>ErrorResponse</code>	<code>message: String</code> , <code>code: int</code>	Error details for failed requests

4. Algorithms/Logic

CSV Upload and Visualization Flow:

```
User uploads CSV file via frontend
→ Frontend sends file to `/api/upload` (POST)
→ Backend (`FileUploadController`) receives file
  → Passes file to `CSVProcessor`
    → Parses CSV, aggregates data (e.g., count by category)
  → Returns `ChartData` JSON
→ Frontend receives data, renders chart using Chart.js
```

5. Error Handling

Scenario	Handling/Response
Invalid/empty file upload	Return 400 with error message
Malformed CSV	Return 422 with parsing error message
Internal server error	Return 500 with generic error message
Unsupported file type	Return 415 with error message

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